Aggregate SQL Functions

The Aggregate Functions in SQL perform calculations on a group of values and then return a single value. Following are a few of the most commonly used Aggregate Functions:

Function	Description
SUM()	Used to return the sum of a group of values.
COUNT()	Returns the number of rows either based on a condition, or without a condition.
AVG()	Used to calculate the average value of a numeric column.
MIN()	This function returns the minimum value of a column.
MAX()	Returns a maximum value of a column.
FIRST()	Used to return the first value of the column.
LAST()	This function returns the last value of the column.

Let us look into each one of the above functions in depth. For your better understanding, I will be considering the following table to explain to you all the examples.

StudentID	StudentName	Marks
1	Sanjay	64
2	Varun	72
3	Akash	45
4	Rohit	86
5	Anjali	92

SUM()

Used to return a total sum of numeric column which you choose.

Syntax:

```sql
SELECT SUM(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the sum of marks of all students from the Students table.

```sql
SELECT SUM(Marks)
FROM Students;
```

Output:

```
359
```

COUNT()

Returns the number of rows present in the table either based on some condition or without any condition.

Syntax:

```
SELECT COUNT(ColumnName)
FROM TableName
WHERE Condition;
```

Example:

Write a query to count the number of students from the Students table.

```
SELECT COUNT(StudentID)
FROM Students;
```

Output:

```
5
```

Example:

Write a query to count the number of students scoring marks > 75 from the Students table.

```
SELECT COUNT(StudentID)
FROM Students
WHERE Marks >75;
```

Output:

```
2
```

AVG()

This function is used to return the average value of a numeric column.

Syntax:

```
SELECT AVG(ColumnName)
FROM TableName;
```

Example:

Write a query to calculate the average marks of all students from the Students table.

```
SELECT AVG(Marks)
FROM Students;
```

Output:

```
71.8
```

MIN()

Used to return the minimum value of a numeric column.

Syntax:

```sql
SELECT MIN(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the minimum marks out of all students from the Students table.

```sql
SELECT MIN(Marks)
FROM Students;
```

Output:

MAX()

Returns the maximum value of a numeric column.

Syntax:

```sql
SELECT MAX(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the maximum marks out of all students from the Students table.

```sql
SELECT MAX(Marks)
FROM Students;
```

Output:

FIRST()

This function returns the first value of the column which you choose.

Syntax:

```sql
SELECT FIRST(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the marks of the first student.

```sql
SELECT FIRST(Marks)
FROM Students;
```

Output:

LAST()

Used to return the last value of the column which you choose.

Syntax:

```
SELECT LAST(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the marks of the last student.

```
SELECT LAST(Marks)
FROM Students;
```

Output: 92

Scalar SQL Functions

The Scalar Functions in SQL are used to return a single value from the given input value. Following are a few of the most commonly used Aggregate Functions:

Let us look into each one of the above functions in depth.

Function	Description
LCASE()	Used to convert string column values to lowercase
UCASE()	This function is used to convert a string column values to Uppercase.
LEN()	Returns the length of the text values in the column.
MID()	Extracts substrings in SQL from column values having String data type.
ROUND()	Rounds off a numeric value to the nearest integer.
NOW()	This function is used to return the current system date and time.
FORMAT()	Used to format how a field must be displayed.

LCASE()

Used to convert values of a string column to lowercase characters.

Syntax:

```
SELECT LCASE(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the names of all students in lowercase.

```
SELECT LCASE(StudentName)
```

```
FROM Students;
```

Output:

```
sanjay
varun
akash
rohit
anjali
```

UCASE()

Used to convert values of a string column to uppercase characters.

Syntax:

```
SELECT UCASE(ColumnName)
FROM TableName;
```

Example:

Write a query to retrieve the names of all students in lowercase.

```
SELECT UCASE(StudentName)
FROM Students;
```

Output:

```
SANJAY
VARUN
AKASH
ROHIT
ANJALI
```

LEN()

Used to retrieve the length of the input string.

Syntax:

```
SELECT LENGTH(String) AS SampleColumn;
```

Example:

Write a query to extract the length of the student name “Sanjay”.

```
SELECT LENGTH(“Sanjay”) AS StudentNameLen;
```

Output:

```
6
```

MID()

This function is used to extract substrings from columns having string data type.

Syntax:

```
SELECT MID(ColumnName, Start, Length)
FROM TableName;
```

Example:

Write a query to extract substrings from the StudentName column.

```
SELECT MID(StudentName, 2, 3)
FROM Students;
```

Output:

```
anj
aru
kas
ohi
nja
```

ROUND()

This function is used to round off a numeric value to the nearest integer.

Syntax:

```
SELECT ROUND(ColumnName, Decimals)
FROM TableName;
```

Example:

For this example, let us consider the following Marks table in the Students table.

StudentID	StudentName	Marks
1	Sanjay	90.76
2	Varun	80.45
3	Akash	54.32
4	Rohit	72.89
5	Anjali	67.66

Write a query to round the marks to the integer value.

```
SELECT ROUND(Marks)
FROM Students;
```

Output:

```
91
80
54
73
68
```

NOW()

Used to return the current date and time. The date and time are returned in the "YYYY-MM-DD HH-MM-SS" format.

Syntax:

```
1SELECT NOW();
```
Example:

Write a query to retrieve the current date and time.

SELECT NOW();

Output:

NOW()
2019-10-14 09:16:36